

Mohammed Muzammil Enam

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ABOUT

Medical Device and IT equipment conformity Assessment professional with expertise in IEC/EN/CSA/ANSI 60601-1, IEC/EN/CSA 62368-1, IEC/CSA 61010-1, UL 1449, Canadian Standard CSA C22.2 No. 269.3, UL 67, Industrial Control Panels UL 508A, Collateral and Particular standard).

WORK EXPERIENCE:

Nemko USA, Inc (Austin, TX)

April 2024 to Present

Safety ~~Laboratory~~ Engineer at Safety Department.

- Experienced in conducting safety evaluation and compliance assessment of:

Medical device in accordance with IEC 60601-1 standards for Product certification (NRTL), CE and CB Scheme evaluation.

Audio/Video, Information and communication technology equipment in accordance with IEC/UL/CSA/AAMI/EN standards for Product certification (NRTL), CE and CB Scheme evaluation.

Measurement, Control and laboratory equipment in accordance with IEC/UL/CSA/AAMI/EN standards for Product certification (NRTL), CE and CB Scheme evaluation.

- Provide Technical Information services to Manufacturer and prepare Engineering evaluation plan based on applicable standards.
- Conducted comprehensive evaluations of Risk Management files, Usability files, and Software documentations, ensuring compliance with IEC 60950, IEC 62368 and IEC 61010 standards.
- Collaborated with cross-functional safety compliance and Environmental Safety Lab (ESL) team to conduct testing and certification for diverse electronics equipment, including Class I, Class II, and internally powered devices.
- Collaborate with Manufacturer and prepare applicable quotes based on applicable standards and complexity of evaluation.
- Execute Test plan to perform applicable tests by setting up testing equipment & recording data for various testing procedures including, but not limited to: Working Voltage, Dielectric, Current Leakage, Anormal & Fault Conditions, Temperature, Ground Impedance, Ingress Protection, Mold Stress Relief & Mechanical Tests.
- Experience in conducting safety tests for Medical equipment and issuing test reports for the IEC 60601-1.
- Experienced in effective communication of project status and test results to sales and management, ensuring informed decision making and streamlined project execution.

Intertek Testing Services, Na, Inc (Plano, TX)

May 2023 to March 2024

Project Engineer at Industrial Department.

- Evaluated and tested various industrial equipment to ensure electrical safety compliance in regulatory & safety guidelines.

- Prepared Test reports and issue ETL Listing reports, LPC (Limited Production Certification) reports, & CB scheme to the provisions of U.S., Canadian & other international product safety standards.
- Identified, properly documented, & recommend engineering judgments within policy guidelines.
- Determined and performed the requisite tests by setting up testing apparatus & recoding data for various testing procedures including, but not limited to: Working Voltage, Dielectric, Current Leakage, Anormal & Fault Conditions, Temperature, Ground Impedance, Ingress Protection, Mold Stress Relief & Mechanical Tests.
- Responsible for daily progress updates, maintaining schedule, reviewing schematic drawings, & providing feedback.
- Travelled to Client's site and perform evaluation.
- Set up and operated standard test equipment including dielectric strength tester, ground continuity tester, digital multi-meter, oscilloscope, data logger, Regulated AC/DC power supply, Power Analyzers, Leakage current tester and environmental condition chamber, etc.
- Specialized in Testing and Evaluation of Surge Protective Devices, Power strips and Industrial Control Panels.
- Industrial Equipment based on UL 1449, Canadian Standard CSA C22.2 No. 269.3, UL 67, Industrial Control Panels UL 508A, Collateral and Particular standard).
- Perform various Preliminary design review to help manufacturer with safety standard requirement.

Calcutta Electric Supply Corporation (Kolkata, India)

Oct 2020 to Nov 2020

Industrial Internship at BSNL

- Detection of fault and repairs in the underground electrical supply lines using 2 lamp method.
- Replacement of old underground Pillar Box to avoid short circuit risks and minimize power loss.
- Installation of new overhead electrical supply line for domestic usage.

Bharat Sanchar Nigam Limited (Kolkata, India)

Jun 2020 to Aug 2020

Industrial Internship at BSNL

- Vocational Training in Advanced Telecom at BSNL - a government owned telecommunication service provider.
- Radio frequency rectification of slow / unstable Internet Speed Problem.
- Practical hands-on session on Optical Fiber Cable and Consumer products like routers.
- Gained hands on experience on the working of Telecommunication Network and Broadband.

EDUCATION

M.S in Electrical Engineering

California State Polytechnic University, Pomona, CA

B.E in Electronics and Telecommunication Engineering

Sathyabama University, Chennai, India

PUBLICATION

Apr 2020: Published Project: Gradient Resource Allocation on LTE based MIMO OFDMA. Researched on Gradient Allocation Method and implemented MIMO OFDMA using MATLAB where the increase of power allocation through relay stations has helped the secondary users in rural areas to get the maximum networks.: [weblink](#)

ACADEMIC PROJECTS

May 2023: Variable Power Supply AC/DC Circuit at 3A current with Overload Protection

Designed and simulated a compact variable power supply using a voltage Integrated circuit regulator with overload protection to power a Unmanned Aerial Vehicle drones during its flight time to monitor electrical powerlines.: [weblink](#)

Dec 2022: Simple Variable Power Supply Circuit with Output 0 – 30V at 2A current

Designed and simulated a variable power supply circuit using LT Spice simulation to achieve 0-30V variable voltage measuring over 2.2A at the output. This circuit design was further used to charge the battery of the Unmanned Aerial Vehicle (UAV) drones to monitor electrical powerlines.

Aug 2022: Project Case Study: Fault Tree Analysis

Computed the reliability, Mean Time between Failure of each component and calculated the overall failure rate. Developed an expression for the overall failure probability and researched root cause analysis and risk analysis based on the component failure probabilities.: [weblink](#)

Oct 2021: CMOS Inverting Buffer as Output Driver for a Chip

Designed CMOS logic Inverters using LT Spice. Simulation. Evaluated Average Power Consumption, Maximum Data Rate and Driving Strength. Demonstrated package effects and temperature effects under different input values.: [weblink](#)

Sep 2021: CMOS Inverting Buffer

Designed a CMOS Inverting buffer using LT Spice Simulation. Evaluated DC Transfer Function, Comparison of Noise Margin, Propagation Delay, Rise and Fall time, and Inverter connection failure.: [weblink](#)

Aug 2015: Microwave, Optical and Simulation Laboratory

Researched on Horn antenna – Gain and directional characteristics, calibrated and monitored the performance accuracy of Microwave component, three port Circulator, Mono mode and Multimode optical fibers, using electronic equipment. Maintained the data spread sheet for further analysis.

Aug 2014: Electronic and Electrical Laboratory

Designed, tested and verified biasing circuit, Half wave and full wave rectifier, series and shunt voltage regulator connecting electronic equipment such a resistor, transistor, Diode and Capacitor to check the output waveform regulating AC/DC power supply and CRO. Designed and simulated electronic circuit in LabVIEW using arrays, loops, structures, sampling aliasing and quantization.

ACADEMIC COURSES

VLSI Circuit Design Lectures and Lab	Programming in C lecture & Lab	Programming In MATLAB	Applied Numerical Methods	Analog Integrated Circuit Lecture & Lab
Object Oriented Programming Lectures & Lab	Network Analysis and Synthesis	Performance Reliability Analysis	Cryptography and Network Security	Microcontroller and Embedded System Lab
Engineering Graphics	Electronic Circuit I & II	Network Security	Nano Electronics	Computer Network

KNOWLEDGE & SKILLS

Laboratory Equipment: AC Supply Source, Oscilloscope, Digital Multimeter, Power analyzer, Electronic data logger, Thermocouple, Fluke Analyzer, Ambient logger, thermocouple (building/application), sensors (Current, voltage, power, flow, speed and temperature).

Proficient in MATLAB, COMSOL, MINITAB, LabVIEW, LTSPICE, Electrical AUTOCAD, EDA, MULTISIM, SIMULINK, PCB, C programming.

Linux, Windows, Ubuntu, Android, SharePoint, MS Office Suite, PowerPoint.

REFERENCES

Available upon request